Q1: Solve the following linear programming problem using the Solver add-in in Microsoft Excel:

a)

$$
\begin{aligned}
\text{Minimize} \quad z = &-0.095x_1 - 0.112x_2 - 0.105x_3 - 0.119x_4 - 0.117x_5 - 0.132x_6 \\
&- 0.105x_7 - 0.109x_8 - 0.055x_9 - 0.051x_{10} \\
\text{subject to} \quad & x_1 + x_2 + x_{11} = 140 \\
& x_3 + x_4 + x_{12} = 160 \\
& x_5 + x_6 + x_{13} = 120 \\
& x_1 + x_7 + x_{14} = 230 \\
& x_3 + x_4 + x_8 + x_{15} = 220 \\
& x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{16} = 600 \\
& x_i \geq 0, \quad i = 1, 2, \ldots, 16.
\end{aligned}
$$

b)

$$
\begin{aligned}
\text{Maximize} \quad & z = x_1 + x_2 \\
& \qquad - 0.105x_7 - 0.109x_8 - 0.055x_9 - 0.051x_{10} \\
\text{subject to} \quad & Ax \leq b,
\end{aligned}
$$

where

$$
A = \begin{bmatrix} 0.1 & 1 \\ 0.2 & 1 \\ 0.4 & 1 \\ 0.6 & 1 \\ 0.8 & 1 \\ 1.0 & 1 \\ 1.2 & 1 \\ 1.4 & 1 \\ 1.6 & 1 \\ 1.8 & 1 \\ 2.0 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad b = \begin{bmatrix} 1.00 \\ 1.01 \\ 1.04 \\ 1.09 \\ 1.16 \\ 1.25 \\ 1.36 \\ 1.49 \\ 1.64 \\ 1.81 \\ 2.00 \end{bmatrix}.
$$

Solution

a)

Methodology

with slack values $x_{14} = 110$ and $x_{11} = x_{12} = x_{13} = x_{15} = x_{16} = 0$. All six equalities are satisfied exactly, so every structural constraint is binding.

Optimal solution

Variable	Value	Variable	Value	Variable	Value
x_1	0	x_7	120	x_{13}	0
x_2	140	x_8	60	x_{14}	110
x_3	0	x_9	0	x_{15}	0
x_4	160	x_{10}	0	x_{16}	0
x_5	0	x_{11}	0		
x_6	120	x_{12}	0		

Table 1: optimal decision-variable values.

$$\boxed{z^\star = -69.700}$$

Constraint	LHS	Sign / RHS	Slack	Status
$x_1 + x_2 + x_{11}$	140	$= 140$	0	Binding
$x_3 + x_4 + x_{12}$	160	$= 160$	0	Binding
$x_5 + x_6 + x_{13}$	120	$= 120$	0	Binding
$x_1 + x_7 + x_{14}$	230	$= 230$	0	Binding
$x_3 + x_4 + x_8 + x_{15}$	220	$= 220$	0	Binding
$\sum_{i=1}^{10} x_i + x_{16}$	600	$= 600$	0	Binding

Table 2: constraint status at the optimum.

Solver Parameters settings

- Set Objective: `$B$34` $\rightarrow$ Min
- By Changing Variable Cells: `$B$16:$B$31` ($x_1, \ldots, x_{16}$)
- Constraints: `$B$38:$B$43` = `$D$38:$D$43` (the six equalities)
- Make Unconstrained Variables Non-Negative: on.The formulation requires $x_i \geq 0$ for all i.
- Solving Method: Simplex LP

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b)

Methodology

Optimal solution

Variable	Value
x_1	0.45
x_2	0.80

Table 3: optimal decision-variable values.

$$\boxed{z^\star = 1.250}$$

Solver Parameters settings

- Set Objective: `$B$14` $\rightarrow$ Max
- By Changing Variable Cells: `$B$10:$B$11` (x_1, x_2)
- Constraints: `$B$18:$B$28` $\leq$ `$D$18:$D$28` (the eleven rows of $Ax \leq b$)

Constraint	LHS	RHS	Slack	Status
$L_1 :\ 0.1x_1 + x_2$	0.845	1.00	0.155	Non-binding
$L_2 :\ 0.2x_1 + x_2$	0.890	1.01	0.120	Non-binding
$L_3 :\ 0.4x_1 + x_2$	0.980	1.04	0.060	Non-binding
$L_4 :\ 0.6x_1 + x_2$	1.070	1.09	0.020	Non-binding
$L_5 :\ 0.8x_1 + x_2$	1.160	1.16	0.000	Binding
$L_6 :\ 1.0x_1 + x_2$	1.250	1.25	0.000	Binding
$L_7 :\ 1.2x_1 + x_2$	1.340	1.36	0.020	Non-binding
$L_8 :\ 1.4x_1 + x_2$	1.430	1.49	0.060	Non-binding
$L_9 :\ 1.6x_1 + x_2$	1.520	1.64	0.120	Non-binding
$L_{10} :\ 1.8x_1 + x_2$	1.610	1.81	0.200	Non-binding
$L_{11} :\ 2.0x_1 + x_2$	1.700	2.00	0.300	Non-binding

Table 4: constraint status at the optimum

- Make Unconstrained Variables Non-Negative: off.The problem states no sign restriction, so the variables are free.
- Solving Method: Simplex LP

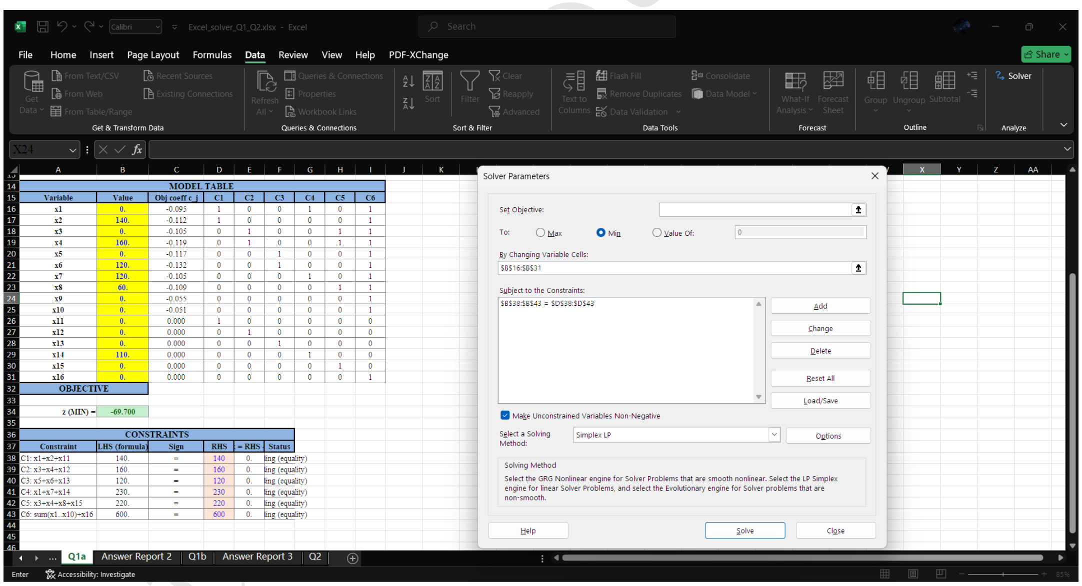

Figure 1: Excel's sheet for Q1-a

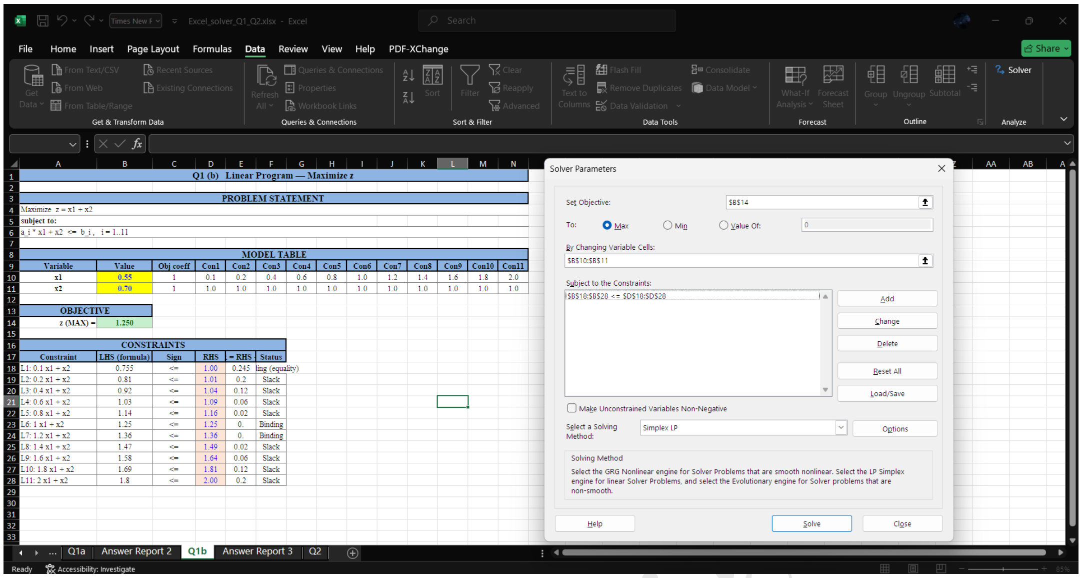

Figure 2: Excel's sheet for Q1-b

Q2: Solve the following nonlinear programming problem using the Solver add-in in Microsoft Excel:

$$
\begin{aligned}
\text{Minimize} \quad f =& (x_1 - 10)^2 + 5(x_2 - 12)^2 + 3(x_4 - 11)^2 + 10x_5^6 + 7x_6^2 \\
& + x_7^4 - 4x_6x_7 - 10x_6 - 8x_7 \\
\text{subject to} \quad g_1 =& 2x_1^2 + 3x_2^4 + x_3 + 4x_4^2 + 5x_5 \leq 127 \\
g_2 =& 7x_1 + 3x_2 + 10x_3^2 + x_4 - x_5 \leq 282 \\
g_3 =& 23x_1 + x_2^2 + 6x_6^2 - 8x_7 \leq 196 \\
g_4 =& 4x_1^2 + x_2^2 - 3x_1x_2 + 2x_3^2 + 5x_6 - 11x_7 \leq 0
\end{aligned}
$$

Solution

Methodology

Special role of x_3.

Since x_3 is absent from the objective, its optimal value is determined entirely by the constraints. At the optimum the constraints g_1 and g_4 are active (binding). From the Karush–Kuhn–Tucker stationarity condition for x_3 with multipliers λ_1 (for g_1) and λ_4 (for g_4),

With the optimal multipliers ($\lambda_1 \approx 1.136$, $\lambda_4 \approx 0.397$) this gives $x_3 \approx -0.7151$, a *negative* value. The Evolutionary engine discovers this value through its population-based search rather than through explicit gradient computation, but the underlying mathematical relationship still holds. This also explains why the non-negativity option must be switched off.

Variable	Value
x_1	2.624706
x_2	1.967182
x_3	−0.358232
x_4	4.312356
x_5	−0.564546
x_6	0.972624
x_7	1.549149

Table 5: optimal decision-variable values.

Optimal solution

$$\boxed{f^\star \approx 680.9297}$$

Constraint	LHS	RHS	Slack	Status
g_1	126.9394	127	0.0000	Binding (active)
g_2	28.3347	282	253.6653	Non-binding
g_3	50.6209	196	145.3791	Non-binding
g_4	−0.1534	0	0.1534	Binding (active)

Table 6: constraint status at the optimum.

Solver Parameters settings

– Set Objective: `$B$22` $\to$ Min

– By Changing Variable Cells: `$B$13:$B$19` $(x_1, \ldots, x_7)$

– Constraints:

  – `$B$26:$B$29` $\leq$ `$D$26:$D$29` $(g_1, \ldots, g_4)$
  – `$B$13:$B$19` ≥ -10 (lower variable bounds for Evolutionary)
  – `$B$13:$B$19` ≤ 15 (upper variable bounds for Evolutionary)

– Make Unconstrained Variables Non-Negative: off. The variables are free, and the optimum has $x_3 < 0$ and $x_5 < 0$.

– Solving Method: Evolutionary

– Initial variable values: $x^{(0)} = (2, 2, 0, 4, 0, 1, 1.5)$, a feasible point placed in the changing cells before solving to seed the initial population.

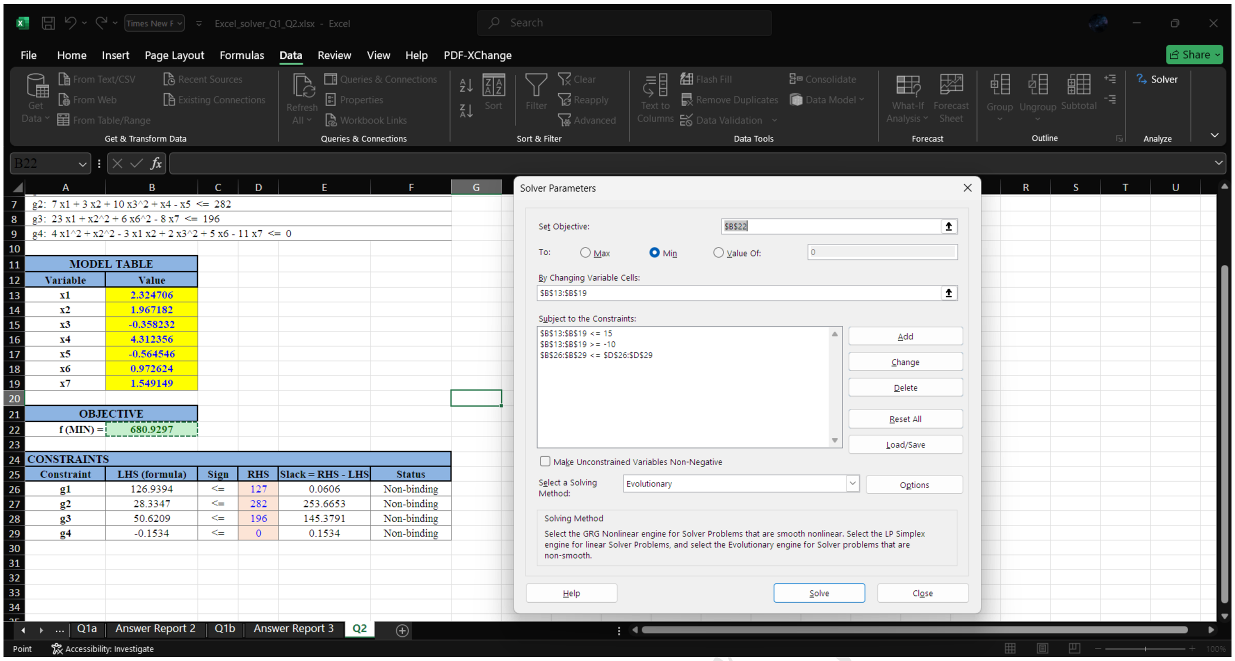

Figure 3: Excel's sheet for Q2

Q3: Find the global minimum points of the following functions using MATLAB software based on genetic algorithm:

a) Griewank Function

$$f(\mathbf{x}) = \frac{1}{4000}\sum_{i=1}^{n} x_i^2 \; - \; \prod_{i=1}^{n} \cos\left(\frac{x_i}{\sqrt{i}}\right) + 1,$$

with bounds

$$-600 \leq x_i \leq 600, \quad i = 1, 2, \ldots, n,$$

and $n = 2$.

b) Holder Table Function

$$f(x) = -\left|\sin(x_1)\cos(x_2)\exp\left(\left|1 - \frac{\sqrt{x_1^2 + x_2^2}}{\pi}\right|\right)\right|$$

c) Ackley Function

$$f(\mathbf{x}) = -a \cdot \exp\left(-b \cdot \sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}\right) - \exp\left(\frac{1}{n}\sum_{i=1}^{n}\cos(cx_i)\right) + a + \exp(1)$$

with bounds

$$-15 \leq x_i \leq 30, \quad i = 1, 2, \ldots, n,$$

and $n = 2$, $a = 20$, $b = 0.2$, and $c = 2\pi$.

d) McCormick Function

$$f(x) = \sin(x_1 + x_2) + (x_1 - x_2)^2 - 1.5x_1 + 2.5x_2 + 1$$

e) Michalewicz Function

$$f(\mathbf{x}) = -\sum_{i=1}^{d} \sin(x_i)\, \sin^{2m}\left(\frac{ix_i^2}{\pi}\right)$$

with $x_i \in [0, \pi]$ for all $i = 1, \ldots, d$, and

$$d = 2, \qquad m = 10.$$

Solution

Note: I used Python instead of MATLAB.

Component	Value used	Why I chose it
Encoding	real vector $\mathbf{x} \in \mathbb{R}^2$	the variables are continuous
Population	200 (300 for (a), 400 for (e))	larger where the box is big or the valleys are narrow
Selection	tournament, size 3	a mild, standard selection pressure
Crossover	SBX, $p_c = 0.9$, $\eta_c = 20$	the usual real-coded settings
Mutation	polynomial, $p_m = 1/n$, $\eta_m = 20$	$\sim$one gene per child; standard η_m
Elitism	keep the best 2	so the best solution is never lost
Termination	cap 500 gens (600 for (a),(e)); early stop after 50 gens (80 for (e)) with no gain $> 10^{-8}$	stop once the best plateaus
Seed	42	so the runs reproduce exactly
Polish	SLSQP, ftol$= 10^{-12}$	gradient refinement of the GA best

Table 7: GA parameter values used for all parts.

a)

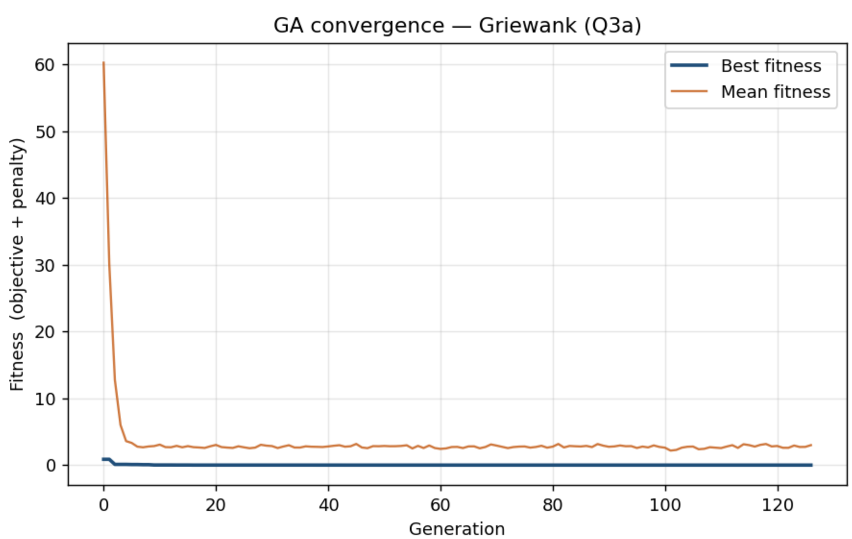

Figure 4: Griewank Function's convergence diagram

SLSQP changed the objective by only $f \approx 2.1 \times 10^{-5}$.

Result

Variable	Value
x_1	0.000001
x_2	0.000000

Table 8: optimal decision-variable values.

$$\boxed{f^\star = 0.000000 \quad \text{at} \quad \mathbf{x}^\star = (0,\, 0)}$$

Best found at generation 76, early stop at 126.

b)

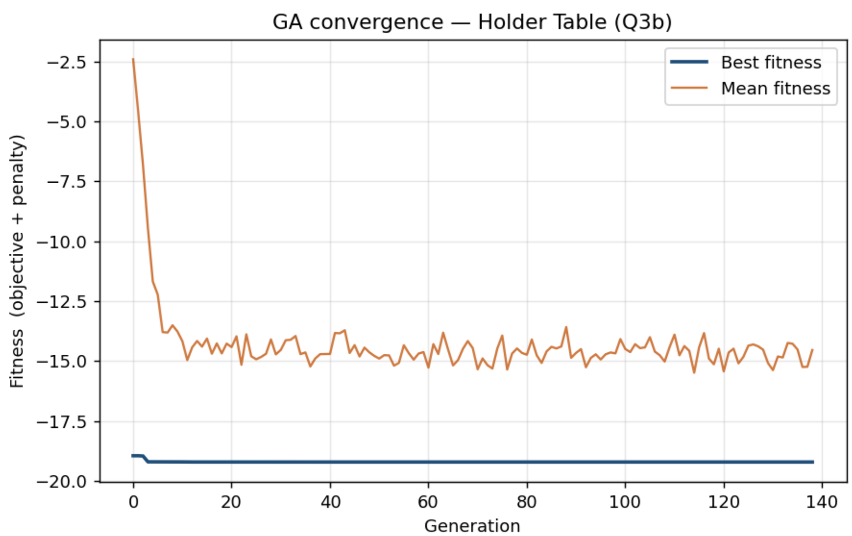

Figure 5: Holder Table Function's convergence diagram

SLSQP changed the objective by only $\approx 6.5 \times 10^{-9}$.

Result

Variable	Value
x_1	-8.055023
x_2	-9.664590

Table 9: optimal decision-variable values.

$$\boxed{f^\star = -19.208503 \quad \text{at} \quad \mathbf{x}^\star = (-8.0550,\, -9.6646)}$$

Best found at generation 88, early stop at 138. All checks pass.

c)

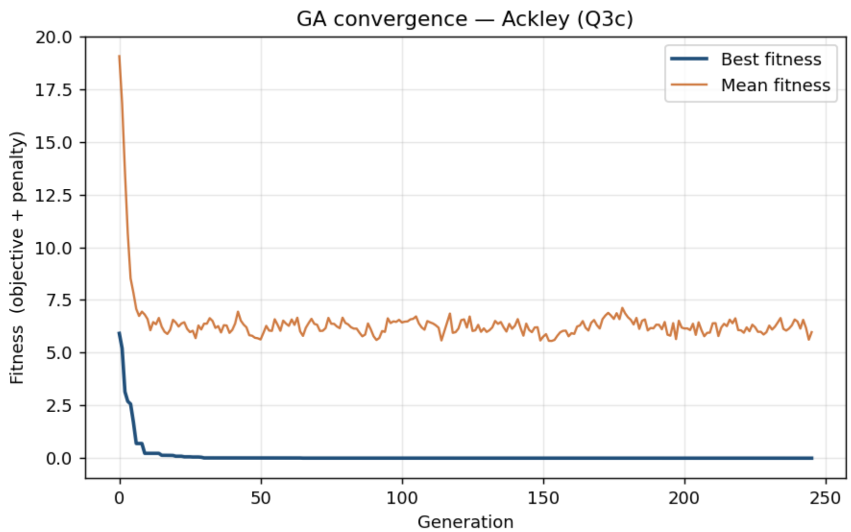

Figure 6: Ackley Function's convergence diagram

SLSQP changed the objective by only $|\Delta f| \approx 2.0 \times 10^{-4}$.

Result

Variable	Value
x_1	0.000000
x_2	-0.000000

Table 10: optimal decision-variable values.

$$\boxed{f^\star = 0.000000 \quad \text{at} \quad \mathbf{x}^\star = (0,\, 0)}$$

Best found at generation 195, early stop at 245. All checks pass.

d)

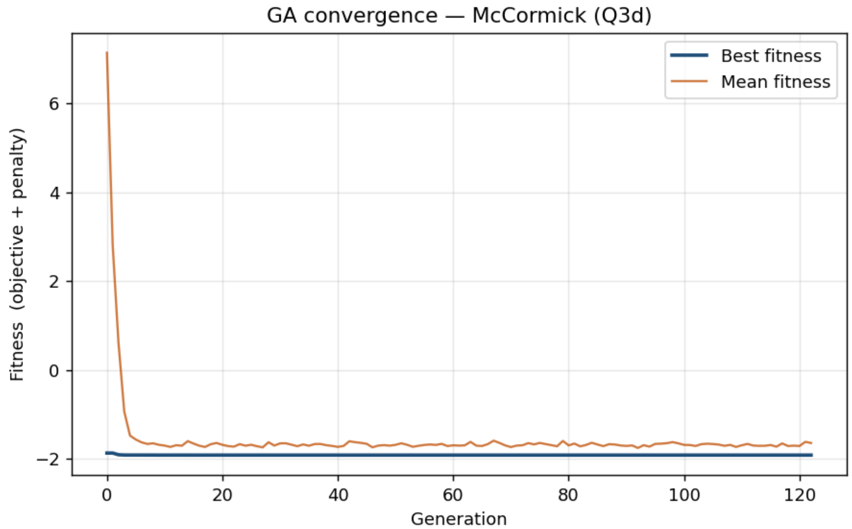

Figure 7: McCormick Function's convergence diagram

SLSQP changed the objective by $\approx 1.0 \times 10^{-8}$.
Result

Variable	Value
x_1	-0.547198
x_2	-1.547198

Table 11: optimal decision-variable values.

$$\boxed{f^{\star} = -1.913223 \;\text{ at }\; \mathbf{x}^{\star} = (-0.5472,\, -1.5472)}$$

Best found at generation 72, early stop at 122. All checks pass.

e)

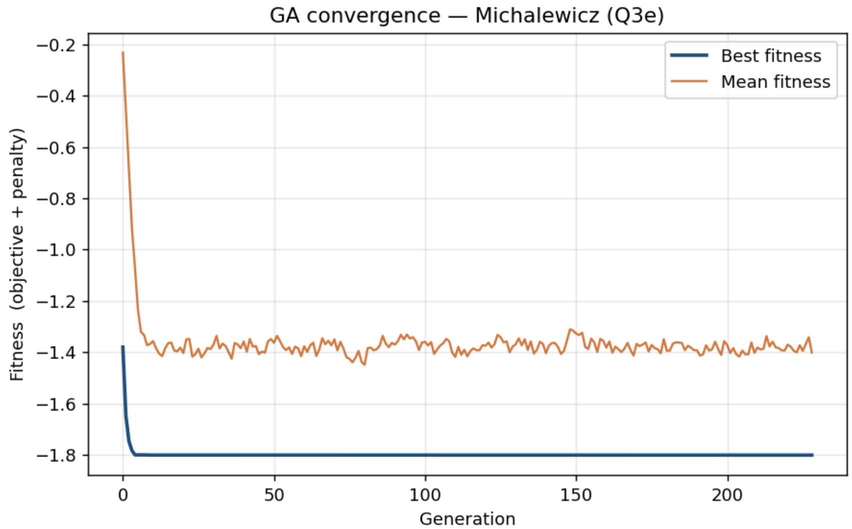

Figure 8: Michalewicz Function's convergence diagram

SLSQP changed the objective by $\approx 1.5 \times 10^{-8}$.

Result

Variable	Value
x_1	2.202906
x_2	1.570796

Table 12: optimal decision-variable values.

$$\boxed{f^\star = -1.801303 \ \text{ at } \ \mathbf{x}^\star = (2.2029,\, 1.5708)}$$

Best found at generation 148, early stop at 228. All checks pass.

Summary

Problem	$f^\star$	$\mathbf{x}^\star$	pop / cap	best / stop gen
Griewank	0.000000	(0, 0)	300 / 600	76 / 126
Holder Table	−19.208503	(−8.0550, −9.6646)	200 / 500	88 / 138
Ackley	0.000000	(0, 0)	200 / 500	195 / 245
McCormick	−1.913223	(−0.5472, −1.5472)	200 / 500	72 / 122
Michalewicz	−1.801303	(2.2029, 1.5708)	400 / 600	148 / 228

Table 13: Obtained values

Q4: Solve the optimal design problem that minimizes the cost function subject to the design constraints for a welded beam, using MATLAB algorithm.

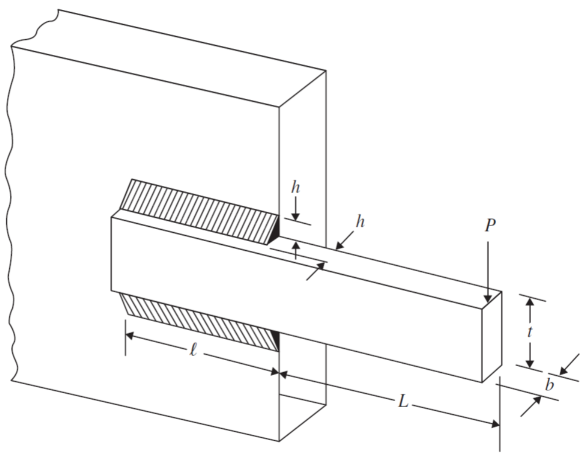

Figure 9: Schematic of the cantilever welded beam

Design vector:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} h \\ l \\ t \\ b \end{bmatrix}$$

where

$$\begin{aligned} \tau &= \text{shear stress in weld,} \\ \sigma &= \text{bending stress in the beam,} \\ P_c &= \text{buckling load on the bar,} \\ \delta &= \text{end deflection of the beam.} \end{aligned}$$

Objective function:

$$f(X) = 1.10471x_1^2x_2 + 0.04811x_3x_4(14.0 + x_2)$$

Constraints:

$$\begin{aligned} g_1(X) &= \tau(X) - \tau_{\max} \leq 0 \\ g_2(X) &= \sigma(X) - \sigma_{\max} \leq 0 \\ g_3(X) &= x_1 - x_4 \leq 0 \\ g_4(X) &= 0.10471x_1^2 + 0.04811x_3x_4(14.0 + x_2) - 5.0 \leq 0 \\ g_5(X) &= 0.125 - x_1 \leq 0 \\ g_6(X) &= \delta(X) - \delta_{\max} \leq 0 \\ g_7(X) &= P - P_c(X) \leq 0 \\ g_8(X) \text{ to } g_{11}(X) &: \ 0.1 \leq x_i \leq 2.0, \quad i = 1, 4 \\ g_{12}(X) \text{ to } g_{15}(X) &: \ 0.1 \leq x_i \leq 10.0, \quad i = 2, 3 \end{aligned}$$

where

$$
\begin{aligned}
\tau(X) &= \sqrt{(\tau')^2 + 2\tau'\tau''\frac{x_2}{2R} + (\tau'')^2} \\
\tau' &= \frac{P}{\sqrt{2}\,x_1x_2}, \qquad \tau'' = \frac{MR}{J}, \qquad M = P\left(L + \frac{x_2}{2}\right) \\
R &= \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1+x_3}{2}\right)^2} \\
J &= 2\left\{\frac{x_1x_2}{\sqrt{2}}\left[\frac{x_2^2}{12} + \left(\frac{x_1+x_3}{2}\right)^2\right]\right\} \\
\sigma(X) &= \frac{6PL}{x_4x_3^2} \\
\delta(X) &= \frac{4PL^3}{Ex_3^3x_4} \\
P_c(X) &= \frac{4.013\sqrt{EG\,(x_3^2x_4^6/36)}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right)
\end{aligned}
$$

Data:

$$P = 6000\,\text{lb}, \quad L = 14\,\text{in.}, \quad E = 30\times10^6\,\text{psi}, \quad G = 12\times10^6\,\text{psi},$$

$$\tau_{\max} = 13,600\,\text{psi}, \quad \sigma_{\max} = 30,000\,\text{psi}, \quad \text{and} \quad \delta_{\max} = 0.25\,\text{in.}$$

Solution

Note: I used Python instead of MATLAB.

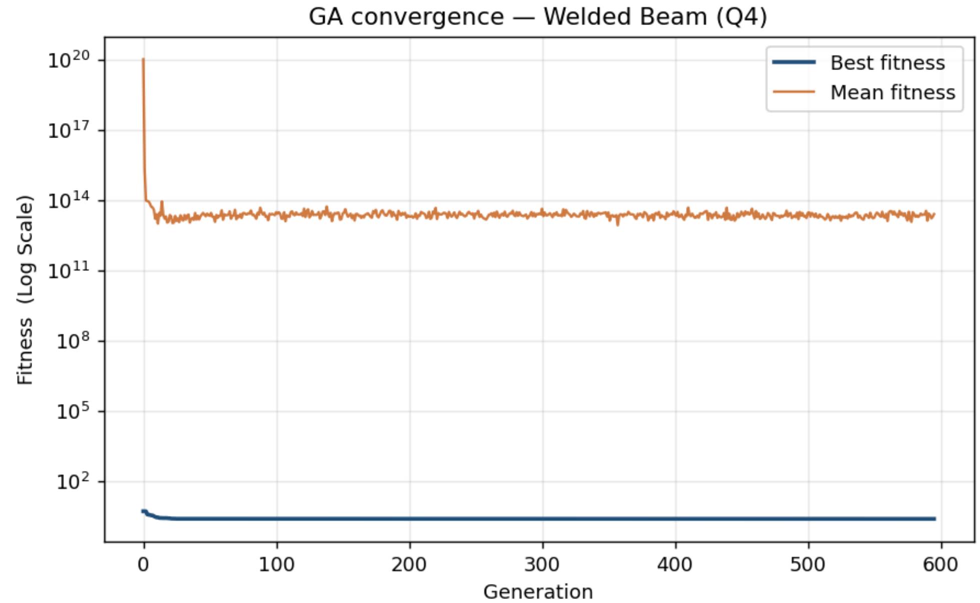

Figure 10: cantilever welded beam's convergence diagram

Before polishing: $f = 2.386766$ at $\mathbf{x} = (0.2456, 6.1964, 8.2695, 0.2457)$. SLSQP local polishing was used to sharpen the variables, driving the active constraints (specifically g_1 for shear stress and g_3 tying weld thickness to beam depth) to be tightly satisfied. This local polish moved the solution noticeably to find the exact boundary of the feasible region ($|\Delta f| \approx 5.809 \times 10^{-3}$).

Result

Variable	Value
x_1 (h)	0.2444
x_2 (l)	6.2175
x_3 (t)	8.2915
x_4 (b)	0.2444

Table 14: Optimal design variables for the welded beam.

$$\boxed{f^\star = 2.380957 \quad \text{at} \quad \mathbf{x}^\star = (0.2444,\, 6.2175,\, 8.2915,\, 0.2444)}$$